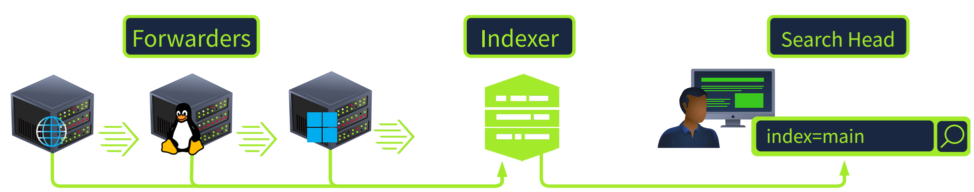
# L2-2.2. Splunk: Setting up a SOC Lab

## Task 1 - Introduction

A few weeks ago, Jasmine, the owner of **Coffely**, reported a data breach where her secret recipe was stolen by an insider from the IT department. Thanks to the Forensics team, the culprit was quickly identified, and the recipe was recovered. Now, Jasmine wants to build an in-house SOC to continuously monitor critical logs and detect threats early. She has enlisted your help to set up **Splunk** locally, integrating logs from a Linux host and the Coffely web server to build a centralized monitoring capability.

## Task 2 - Splunk Deployment Architecture

Before setting up your environment, it is important to understand how Splunk's deployment architecture works. Let's take a look at an overview of the main components that make up the Splunk environment and how they interact in a SOC setting.



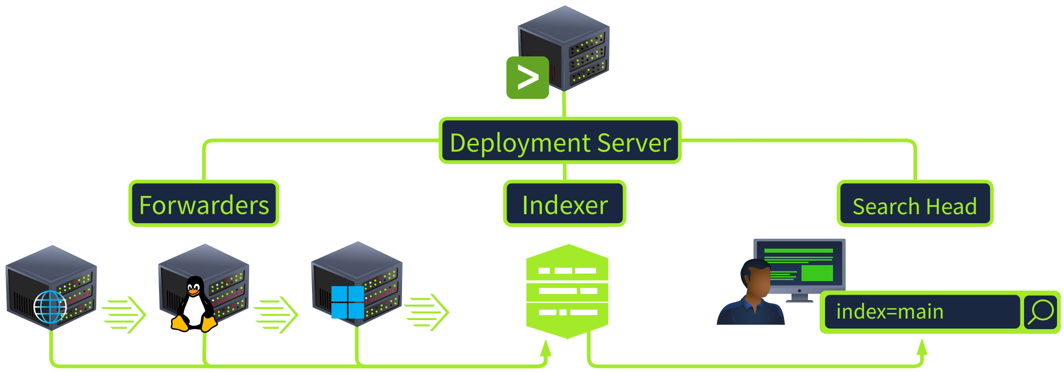

### Architecture Components

**Forwarders: Collect and send raw or filtered data from endpoints and servers to Splunk.**

**Indexers:** Process, parse, transform, and store incoming data to make logs searchable and analyzable.

**Search Head:** Allows analysts to search, investigate, visualize, and correlate data stored by Splunk indexers.

### Deployment Server




Centrally manages and configures large numbers of Splunk forwarders remotely, keeping their configurations synchronized.

**Answer the questions below**

> Which part of the Splunk deployment architecture is responsible for processing incoming data from forwarders?

Indexer

> Part of your client's infrastructure includes hundreds of employee workstations. Which Splunk feature would enable you to remotely manage forwarders on each system?

Deployment Server

## Task 3 - Linux Deployment

If Splunk stops responding at any point, run `/opt/splunk/bin/splunk restart` as `root` and wait for a few minutes. For convenience, the Splunk installer has already been downloaded and placed in the `~/Downloads/splunk` directory.

### Splunk Install

```
ubuntu@coffely:~$ cd Downloads/splunk
ubuntu@coffely:~/Downloads/splunk$ ls
splunk_installer.tgz splunkforwarder.tgzubuntu@coffely:~/Down
loads/splunk/$ sudo su
```

```
root@coffely:~/Downloads/splunk/#
root@coffely:~/Downloads/splunk/# tar xvzf splunk_installer.t
gz -C /opt
splunk/
splunk/splunk-9.0.3-dd0128b1f8cd-linux-2.6-x86_64-manifest
...
```

**Starting Splunk**

```
root@coffely:~/Downloads/splunk/# cd /opt/splunk/bin
root@coffely:/opt/splunk/bin# ./splunk start --accept-license
This appears to be your first time running this version of Sp
lunk.

Splunk software must create an administrator account during s
tartup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in crede
ntials.

Please enter an administrator username: admin
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
...
Waiting for web server at http://127.0.0.1:8000 to be availab
le............... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://coffely:8000
```

**Accessing Splunk:** To access Splunk, open the browser within the VM and go to the address `http://coffely:8000`. If you are connected to the VPN, you can access

Splunk directly in your browser by navigating to `http://MACHINE_IP:8000`. By default, Splunk Web is accessible over port `8000`. Use the credentials you created during the installation process to access the Splunk dashboard.

Answer the questions below

> In which Linux directory, typically used for add-on or third-party software, should Splunk be installed?

/opt

> Which port does the Splunk web interface run on by default?

8000

## Task 4 - Managing Splunk From CLI

Now that you have installed Splunk, it's essential to learn some key commands when interacting with Splunk instances through the CLI. These commands are run from the `/opt/splunk/bin` directory. The `splunk start` command is used to start the Splunk server. This command starts all the necessary Splunk processes, enabling the server to accept incoming data. If the server is already running, this command will have no effect. Splunk can be stopped with `splunk stop`. Restarting Splunk is as easy as running `splunk restart` and is useful when changes have been made to Splunk's configuration.

SplunkStart

```
root@coffely:/opt/splunk/bin# ./splunk start
Splunk> Finding your faults, just like mom.
…
Checking prerequisites...
	Checking http port [8000]: open
	Checking mgmt port [8089]: open
	Checking appserver port [127.0.0.1:8065]: open
	Checking kvstore port [8191]: open
	Checking configuration... Done.
...
```

```
...
The Splunk web interface is at http://coffely:8000
```

#### Further Useful Commands

Use the `splunk enable boot-start` command to configure Splunk to automatically start as a service whenever the system boots. The `splunk status` command is used to check the status of the Splunk server. This command will display information about the current state of the server, including whether it is running or not, and any errors that may be occurring.

SplunkStatus

```
root@coffely:/opt/splunk/bin# ./splunk status
splunkd is running (PID: 45852).
splunk helpers are running (PIDs: 45853 46296 46305 46414 46511 46786).
```

The `splunk add oneshot` command is used to add a single event to the Splunk index. This is useful for testing purposes or for adding individual events that may not be part of a larger data stream.

SplunkOne Shot

```
root@coffely:/opt/splunk/bin# ./splunk add oneshot /path/to/logfile -index yourindex
...
```

The `splunk search` command is used to search for data in the Splunk index. This command can be used to search for specific events, as well as to perform more complex searches using Splunk's search processing language.

SplunkSearch

```
root@coffely:/opt/splunk/bin# ./splunk search coffely
WARNING: Server Certificate Hostname Validation is disabled.
Please see server.conf/[sslConfig]/cliVerifyServerName for details.
```

```
Feb 18 21:09:04 coffley ubuntu: coffely-has-the-best-coffee-i
n-town
Feb 18 13:48:17 coffely ubuntu: COFFELY
Feb 18 13:48:17 coffely ubuntu: COFFELY
....
```

Try out `splunk help` to view all available Splunk CLI commands, their descriptions, and usage syntax.

SplunkHelp

```
root@coffely:/opt/splunk/bin# ./splunk help
Welcome to Splunk's Command Line Interface (CLI).

   Type these commands for more help:

       help [command]             type a command name to acce
ss its help page
       help [object]              type an object name to acce
ss its help page
       help [topic]               type a topic keyword to get
help on a topic
       help commands              display a full list of CLI
commands
       help clustering            commands that can be used t
o configure the clustering setup
       help shclustering          commands that can be used t
o configure the Search Head Cluster setup
       help control, controls     tools to start, stop, manag
e Splunk processes
       help datastore             manage Splunk's local files
ystem use
    ....
```

Answer the questions below

> Using the CLI, what command would you run to search Splunk for the term coffely?

./splunk search coffely

> Which Splunk command would you run to get help with Splunk searches?

./splunk help search

## Task 5 - Splunk Universal Forwarder

The 64-bit version for Linux has been placed in the `~/Downloads/splunk` directory. Navigate back to the `/home/ubuntu/Downloads/splunk` directory. Ensure you're running as root, and run the `tar` command to install the forwarder and move it into the `/opt` directory.

Forwarder Install

```
ubuntu@coffely:~/Downloads/splunk/$ ls
splunk_installer.tgz splunkforwarder.tgzubuntu@coffely:~/Down
loads/splunk/$ sudo su
root@coffely:~/Downloads/splunk/#root@coffely:~/Downloads/spl
unk/# tar xvzf splunkforwarder.tgz -C /opt
splunkforwarder/
splunkforwarder/swidtag/
...
```

### Starting the Forwarder

Navigate to `/opt/splunkforwarder/bin` and use `./splunk start --accept-license` to start the forwarder and enter your new credentials. By default, both Splunk Enterprise and the Splunk Universal Forwarder use port `8089` for management traffic (splunkd). Since both are running on the same machine, this creates a port conflict. When prompted, change the forwarder management port to `8090` so both services can run simultaneously.

Forwarder Start

```
root@coffely:~/Downloads/splunk/# cd /opt/splunkforwarder/bin
root@coffey:/opt/splunkforwarder/bin# ./splunk start --accept
-license
This appears to be your first time running this version of Sp
lunk.
...
...
Please enter an administrator username: admin
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
...
...
Checking prerequisites...
    Checking mgmt port [8089]: not available
ERROR: mgmt port [8089] - port is already bound.  Splunk need
s to use this port.
Would you like to change ports? [y/n]: y
Enter a new mgmt port: 8090
Setting mgmt to port: 8090
The server's splunkd port has been changed.
    Checking mgmt port [8090]: open
Starting splunk server daemon (splunkd)...
Done
```

Answer the questions below

> What is the full command you used to start the forwarder?

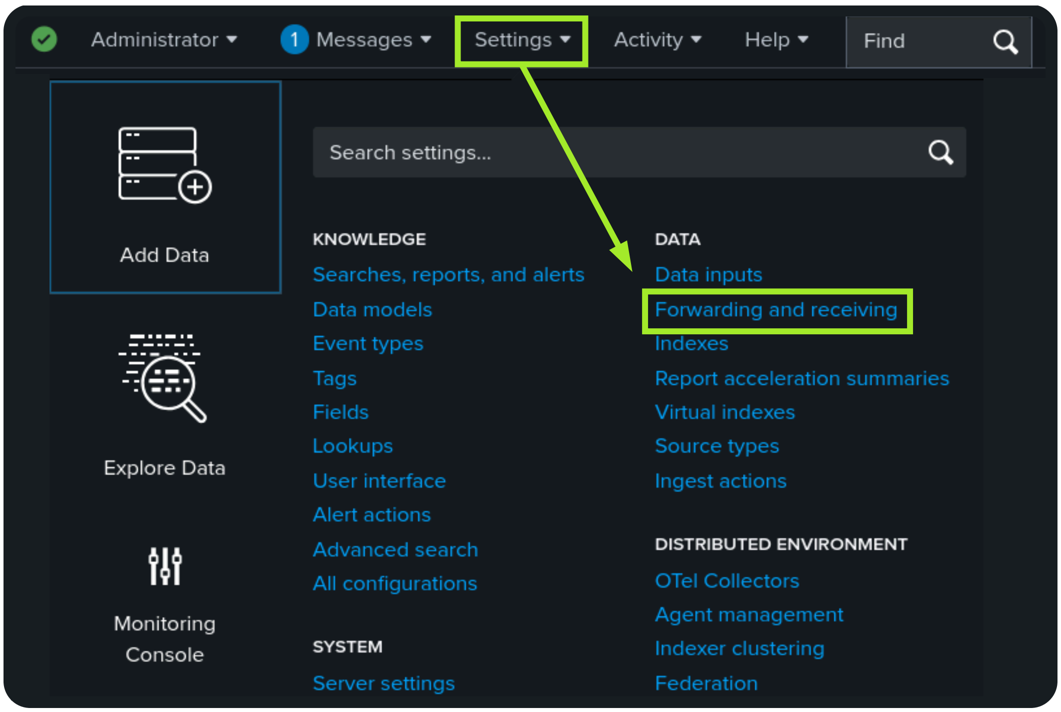
./splunk start --accept-license

> Which port does the Splunk forwarder run on by default?

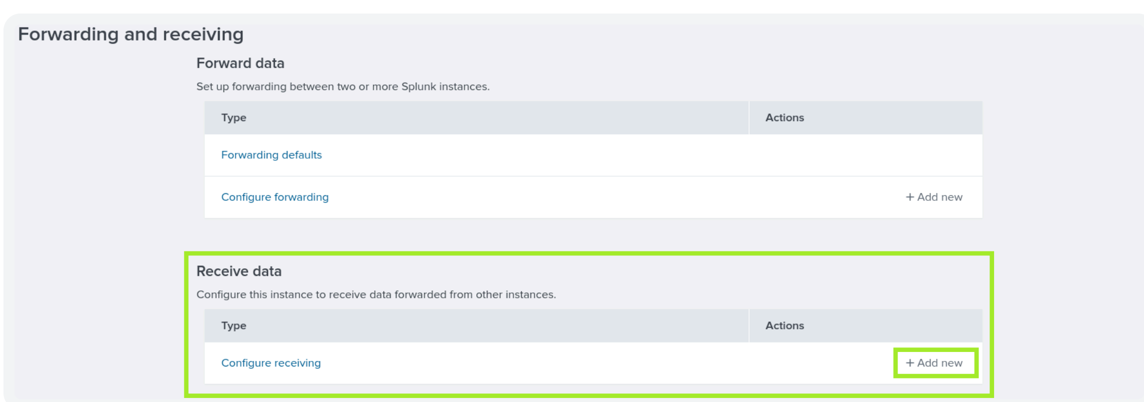
8089

## Task 6 - Configuring the Forwarder

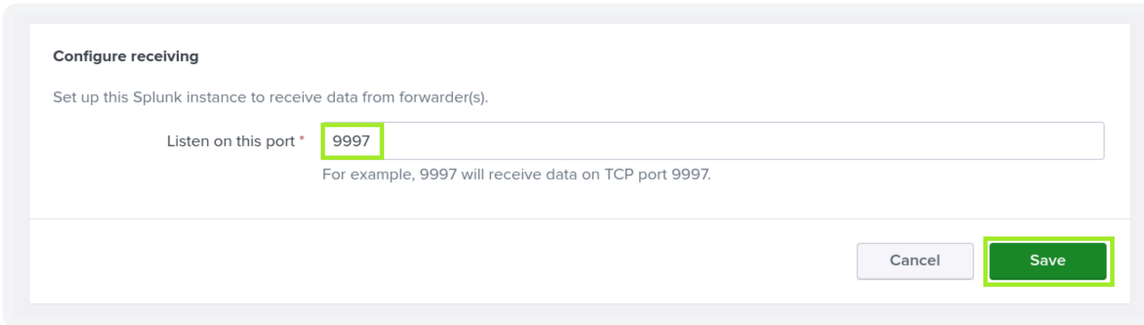
**Splunk Configuration:** Log in to Splunk, head to **Settings**, and click **Forwarding and receiving**.



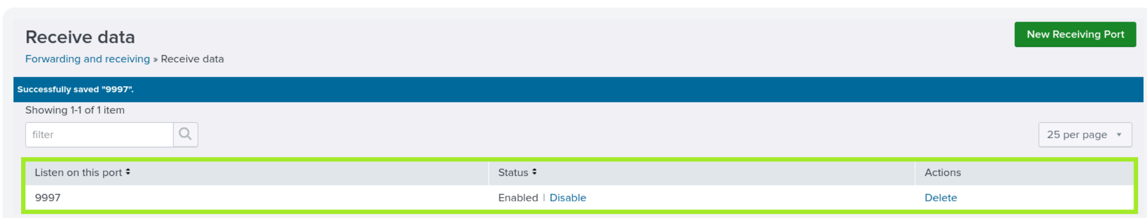

You will see multiple options to configure both forwarding and receiving. We want to receive data from the Linux endpoint, so click `+ Add new` in the **Receive data** section.



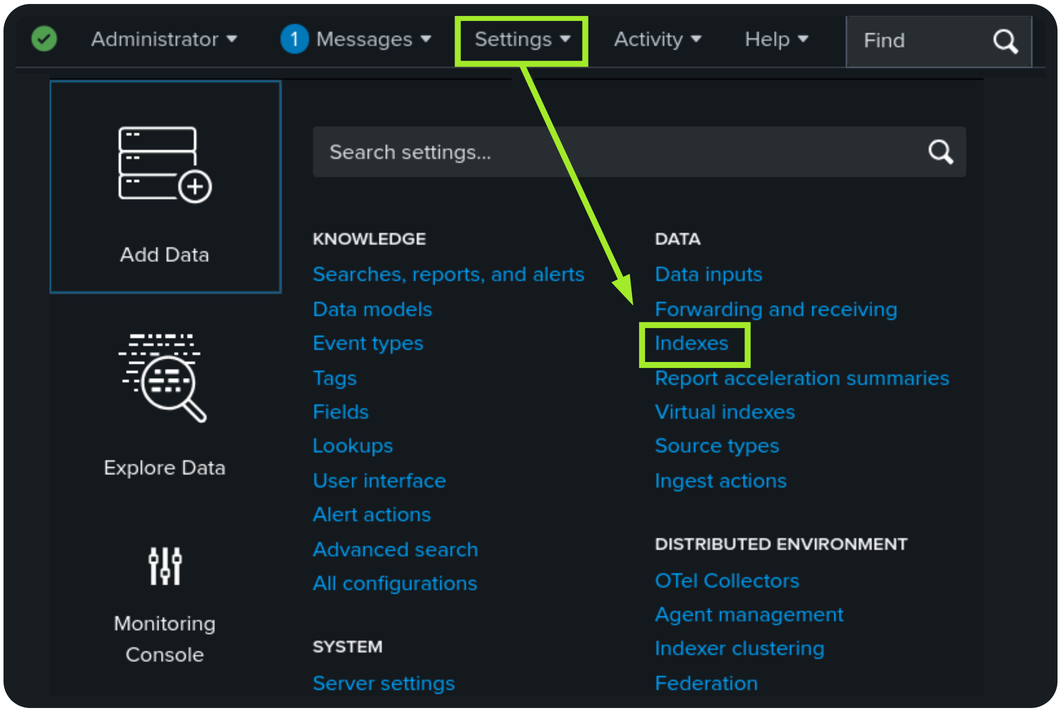

By default, the Splunk instance receives data from the forwarder on port `9997`. It's up to us whether we want to use this port or change it. For now, we will configure our Splunk to start listening on port `9997` and **Save**, as shown below.



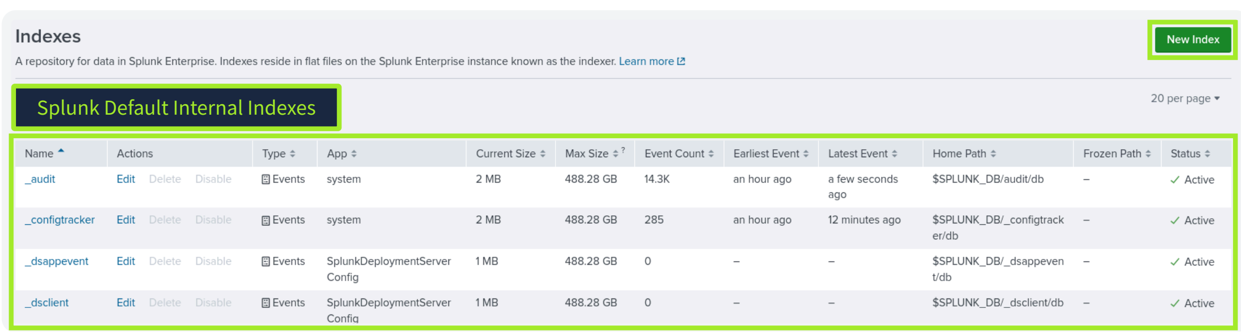

Our listening port `9997` is now enabled and waiting for data. At any time, you can return to this page and disable or delete this configuration.




**Creating an Index:** Head back to **Settings** and click on the **Indexes** option. We will create a new index to store the data we want to receive. If no index is created and specified, Splunk will store received data in the default index `main`.

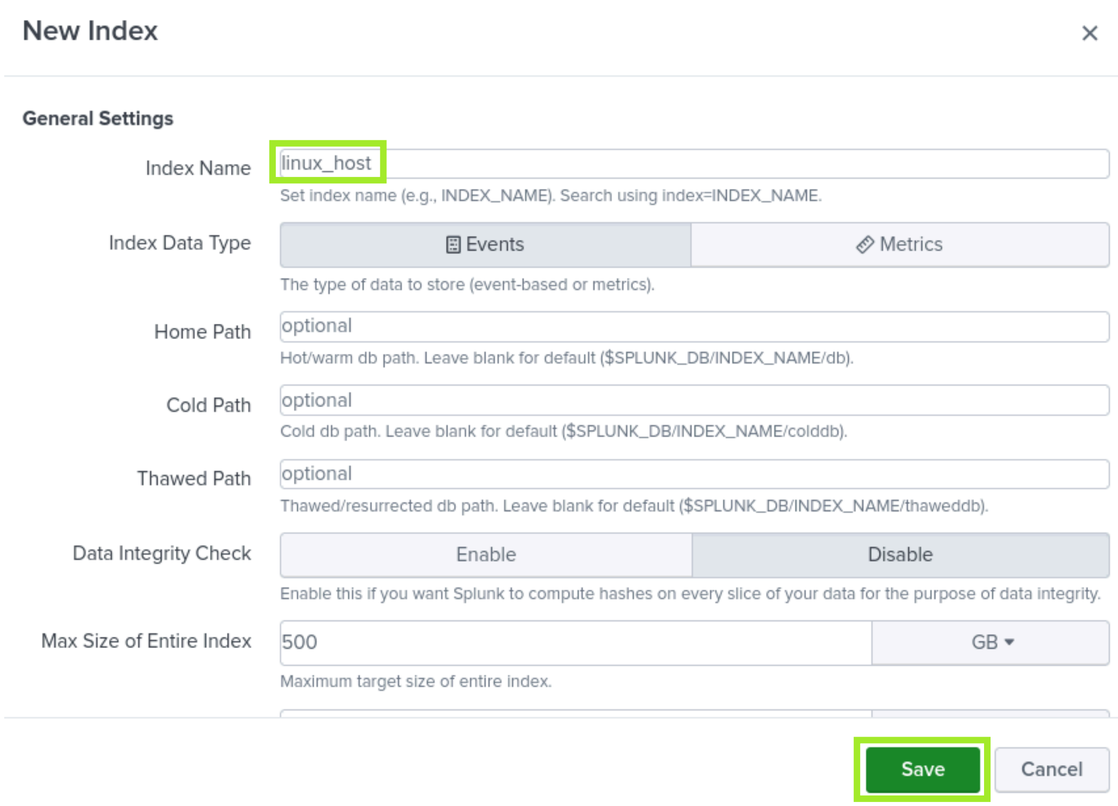



The indexes tab contains all the indexes created by the user or by default. By default, Splunk contains internal logs used to monitor its own operations and performance.  It shows important metadata about the indexes, like size, event count, and home path. Go ahead and click the **New Index** button.




| Name | Actions | Type | App | Current Size | Max Size | Event Count | Earliest Event | Latest Event | Home Path | Frozen Path | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| _audit | Edit Delete Disable | Events | system | 2 MB | 488.28 GB | 14.3K | an hour ago | a few seconds ago | $SPLUNK_DB/audit/db | – | ✓ Active |
| _configtracker | Edit Delete Disable | Events | system | 2 MB | 488.28 GB | 285 | an hour ago | 12 minutes ago | $SPLUNK_DB/_configtracker/db | – | ✓ Active |
| _dsappevent | Edit Delete Disable | Events | SplunkDeploymentServerConfig | 1 MB | 488.28 GB | 0 | – | – | $SPLUNK_DB/_dsappevent/db | – | ✓ Active |
| _dsclient | Edit Delete Disable | Events | SplunkDeploymentServerConfig | 1 MB | 488.28 GB | 0 | – | – | $SPLUNK_DB/_dsclient/db | – | ✓ Active |

Add the index name `linux_host` and keep the default settings for the rest. Go ahead and save your new index.




### Configuring the Forwarder

On the Linux host terminal, add the forward server using the command below.

Add Forward Server

```
root@coffely:/opt/splunkforwarder/bin# ./splunk add forward-server MACHINE_IP:9997
Splunk username: admin
Password:
Added forwarding to: MACHINE_IP:9997.
```

### Linux Log Sources

Linux stores all of its important logs into the `/var/log` directory, as shown below. In our case, we will ingest the `syslog` file into Splunk. All other logs can be ingested using the same method.

Linux Log Sources

```
root@coffely:/opt/splunkforwarder/bin# ls /var/log
README                btmp.1                  gpu-manager.log
speech-dispatcher
Xorg.0.log            cloud-init-output.log   hp
sssd
Xorg.0.log.old        cloud-init.log          journal
syslog
alternatives.log      cloud-init.log.1        kern.log
syslog.1
alternatives.log.1    cups                    kern.log.1
sysstat
amazon                cups-browsed            landscape
ubuntu-advantage.log
apache2               dist-upgrade            lastlog
ubuntu-advantage.log.1
apport.log            dmesg                   lightdm
unattended-upgrades
apport.log.1          dmesg.0                 openvpn
upgrade
apt                   dpkg.log                prime-offload.log
wtmp
auth.log              dpkg.log.1              prime-supported.l
og
auth.log.1            fontconfig.log          private
btmp                  gpu-manager-switch.log  samba
```

Next, use the `./splunk add monitor` command below to add the `syslog` file to our newly created index.

Add Monitor

```
root@coffely:/opt/splunkforwarder/bin# ./splunk add monitor /
var/log/syslog -index linux_host
Added monitor of '/var/log/syslog'.
```

**Inputs:** We can now look at the `inputs.conf` file to check our configuration. Splunk inputs define what data Splunk should collect. You can manually edit `inputs.conf` to specify new data sources or modify existing input settings.

Inputs

```
root@coffely:/opt/splunkforwarder/bin# cat /opt/splunkforwarder/etc/apps/search/local/inputs.conf
[monitor:///var/log/syslog]
disabled = false
index = linux_host
```

### Using the Logger Utility

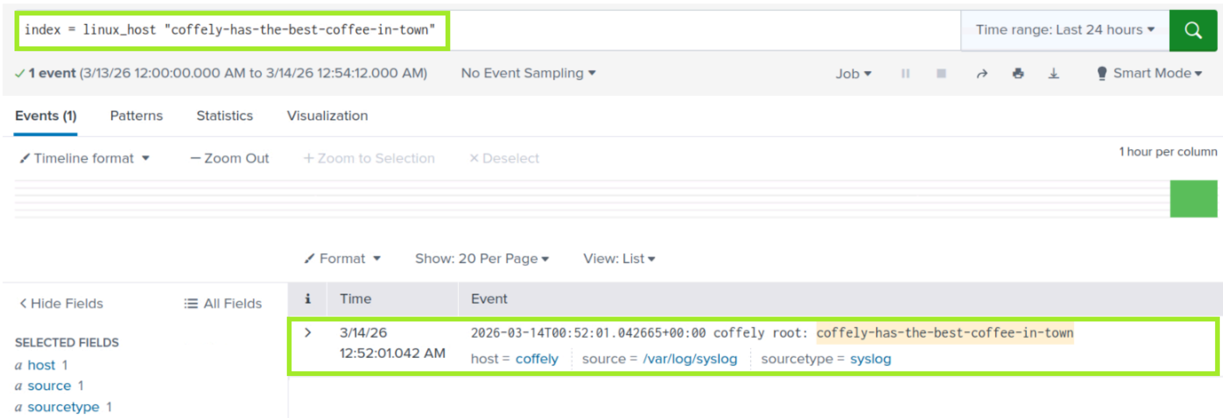
Logger is a built-in command-line tool to add test logs to the `syslog` file. We are already monitoring the `syslog` file and sending all logs to Splunk, so the log generated with the command below will be searchable in our Splunk instance.

Logger Utility

```
root@coffely:/opt/splunkforwarder/bin# logger "coffely-has-the-best-coffee-in-town"
root@coffely:/opt/splunkforwarder/bin# tail -1 /var/log/syslog
2025-11-06T05:37:11.415531+00:00 coffely root: coffely-has-the-best-coffee-in-town
```

To confirm that Splunk is indeed receiving your logs, head to the **Search & Reporting** app in your Splunk instance and search for your test log with the following query:

- `index = linux_host "coffely-has-the-best-coffee-in-town"`




Answer the questions below

> Follow the steps from above, but this time to ingest the /var/log/auth.log file into the linux_host index. What is the value in the sourcetype field?

auth (index = “linux_host” ,check sourcetype

> Create a new user named analyst using the command adduser analyst. Once created, look at the syslog events generated in Splunk related to the user creation activity. What is the value of the process field when the new user is added?

adduser

## Task 7 - Windows Forwarding

**Log Aggregation:** Centralizes logs from multiple systems into one searchable location.

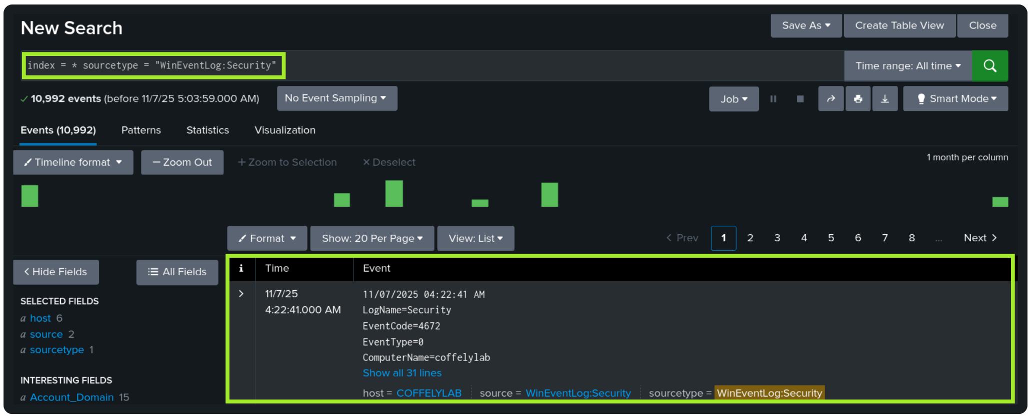
**ecurity Logs:** PowerShell can configure the forwarder to monitor and forward Windows Security logs.

PowerShellAdd Monitor

```
PS C:\> cd "Program Files\SplunkUniversalForwarder\bin"
PS C:\Program Files\SplunkUniversalForwarder\bin> .\splunk.ex
e add monitor C:\Windows\System32\winevt\Logs\Security.evtx
```

```
Added monitor of 'C:\Windows\System32\winevt\Logs\Security.ev
tx'.
```

In **Search & Reporting**, we can now see our newly forwarded Windows Security logs.

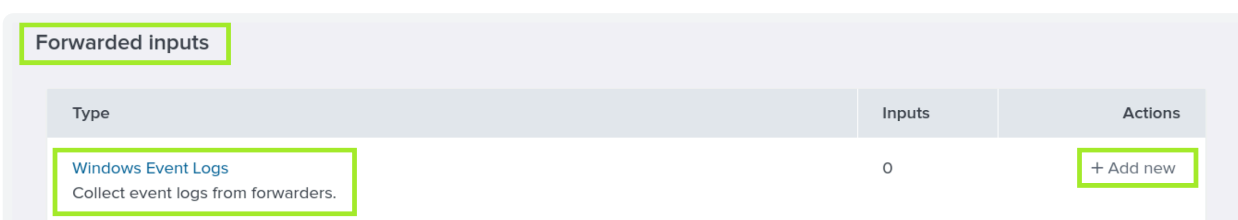



**Deployment Server:** Alternatively, you can turn your Splunk instance into a deployment server(opens in new tab), allowing you to manage other Splunk instances and forwarders remotely. This will give us access to manage our Windows forwarder via the ui.

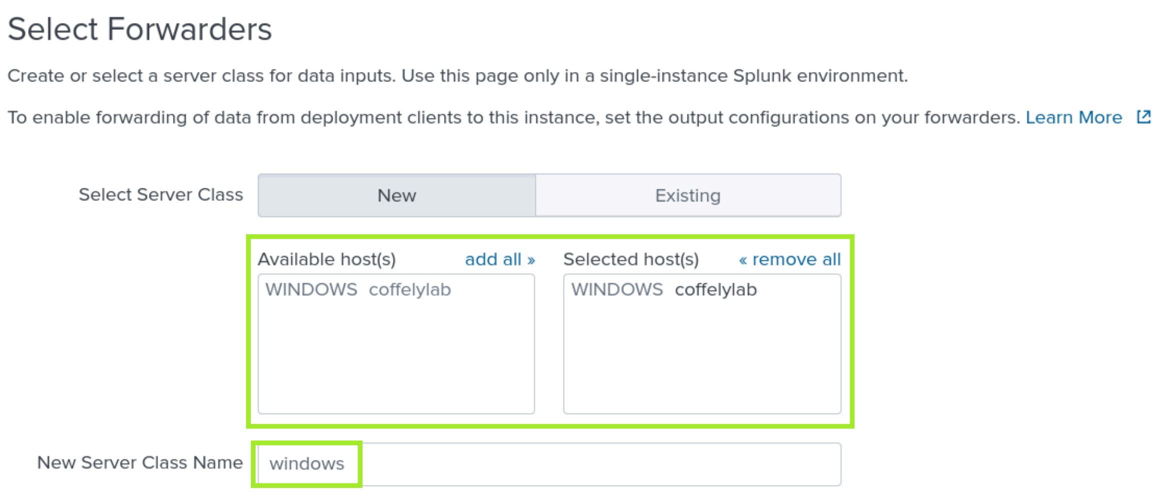
Deployment Server

```
root@coffely:/opt/splunk/bin# ./splunk enable deploy-server
Deployment Server is enabled.root@coffely:/opt/splunk/bin# ./
splunk restart
```

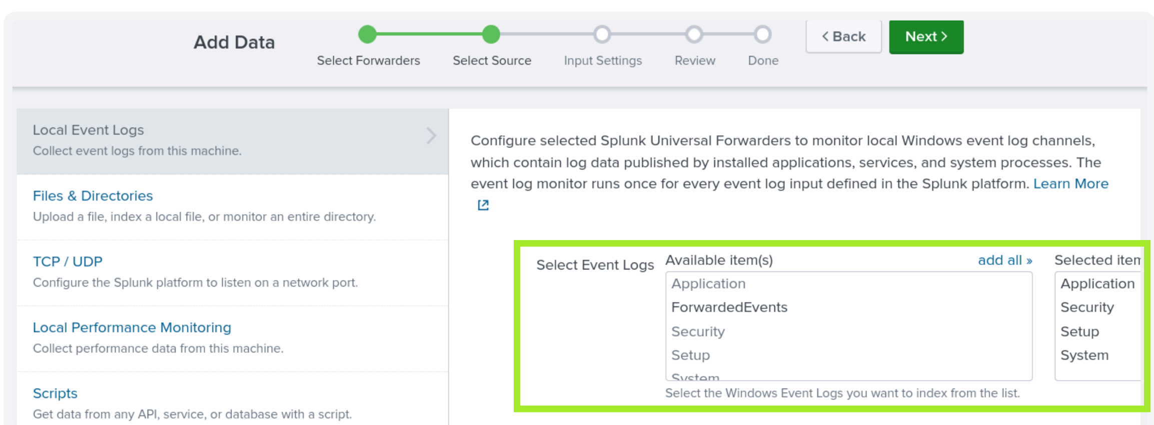
Heading back to **Settings** and **Data Inputs**, scroll down to **Forwarded inputs** and choose `+ Add new` next to **Windows Event Logs**.



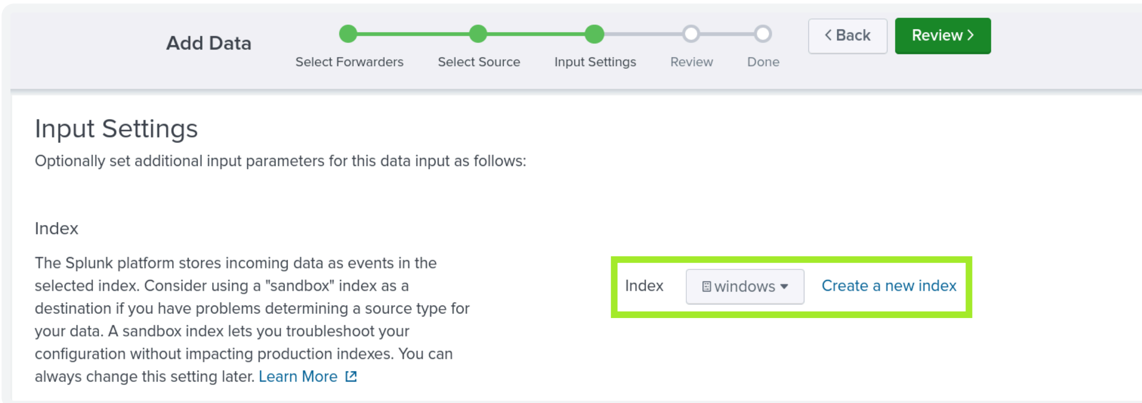
We now see the Windows machine in the Available hosts column. Select it, and give it a new server class name. Server classes are used to define a single instance or group of Splunk deployment clients (Splunk instance, forwarder, indexer, or search head).



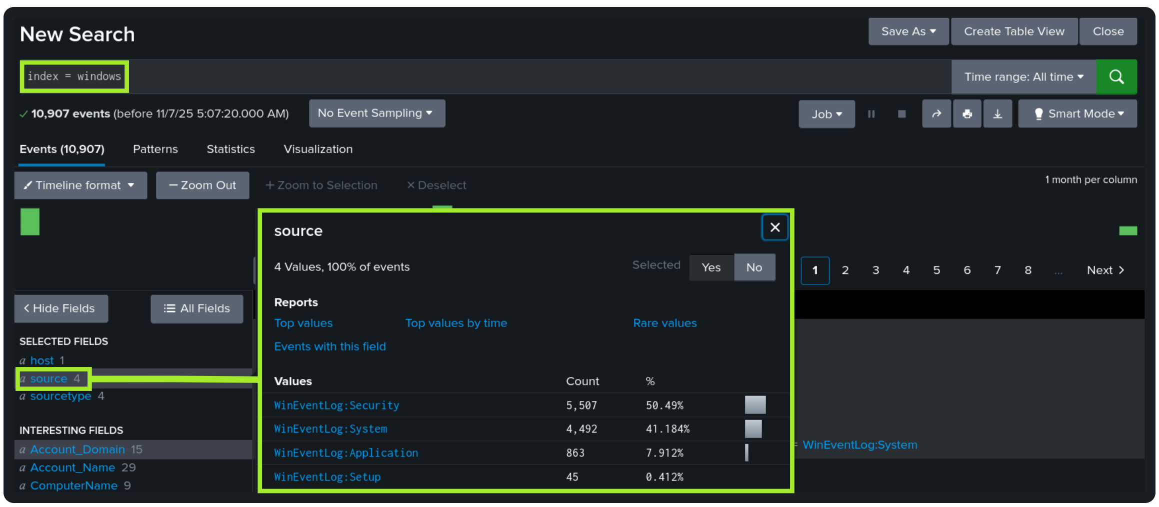

Next, you must choose which log types you want forwarded to Splunk. In this case, Application, Security, Setup, and System have been chosen.




Finally, we can create a new index named `windows`, review our data input, and submit just as we did in the previous steps.




Checking out the newly created `windows` index in the **Search & Reporting** app.



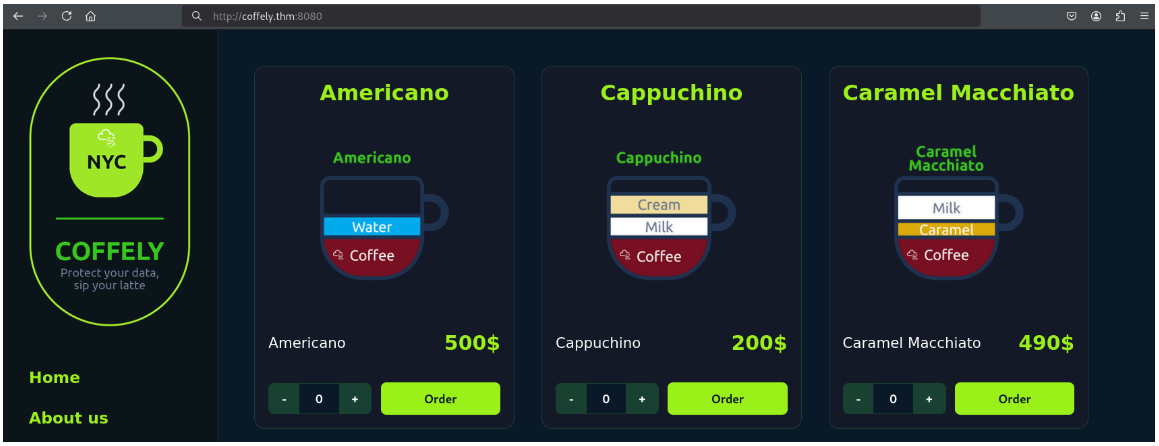

### Answer the questions below

> What command would you use on Linux to enable your Splunk instance as a deployment server?

./splunk enable deploy-server

## Task 8 - Ingesting Web Logs

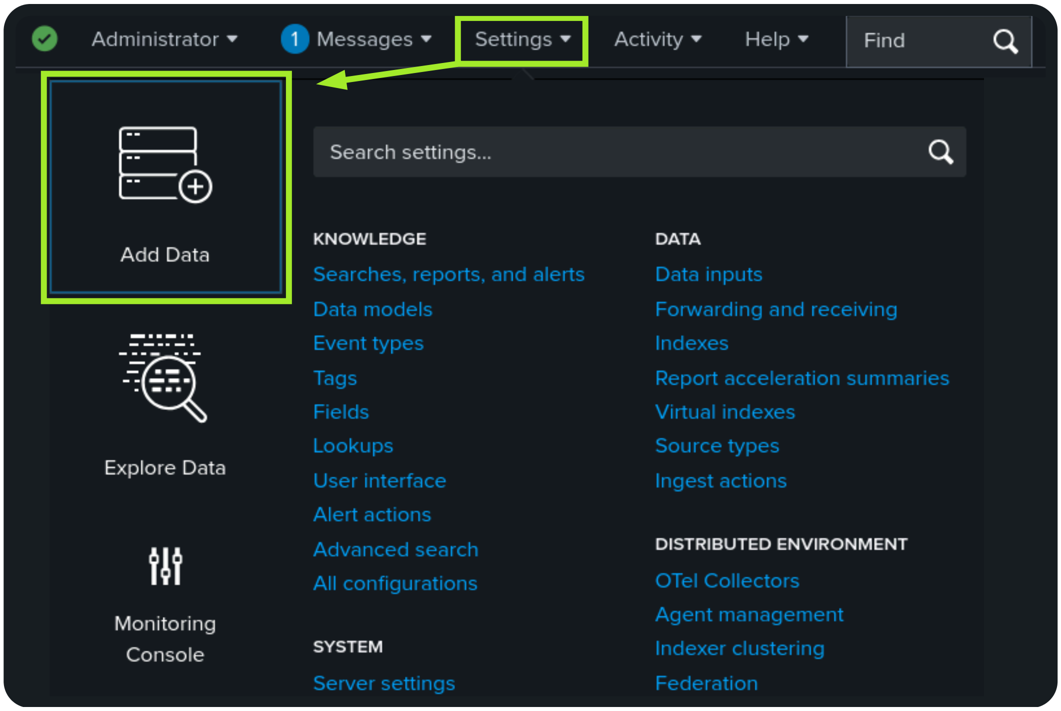
The Linux machine where we have Splunk running also hosts a local copy of their website, which is under development. It can be accessed via `http://coffely.thm:8080` from the VM. You are asked to configure Splunk to receive the web logs from this website to trace the orders and improve coffee sales.



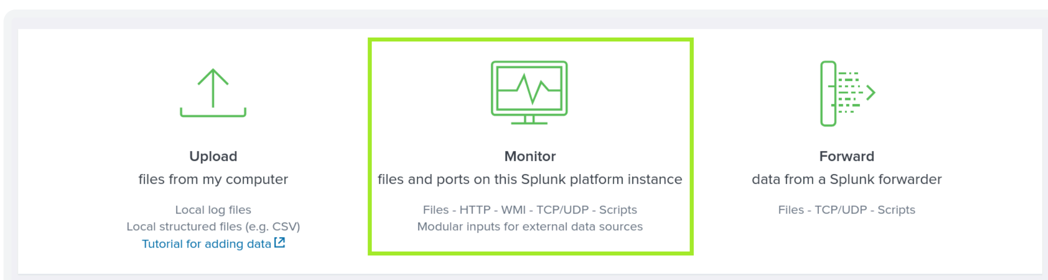

The site will eventually allow users to order coffee online. In the backend, it will track all requests, responses, and orders placed. Let's get started ingesting the web logs into Splunk. This site is currently run on Apache, and we can find the access logs we want to ingest at `/var/log/apache2/access.log`. There are several methods for transferring these logs to Splunk. You can use the command line as we did in the previous tasks, but let's use the Splunk interface to monitor the `access.log` file.

### Adding Data

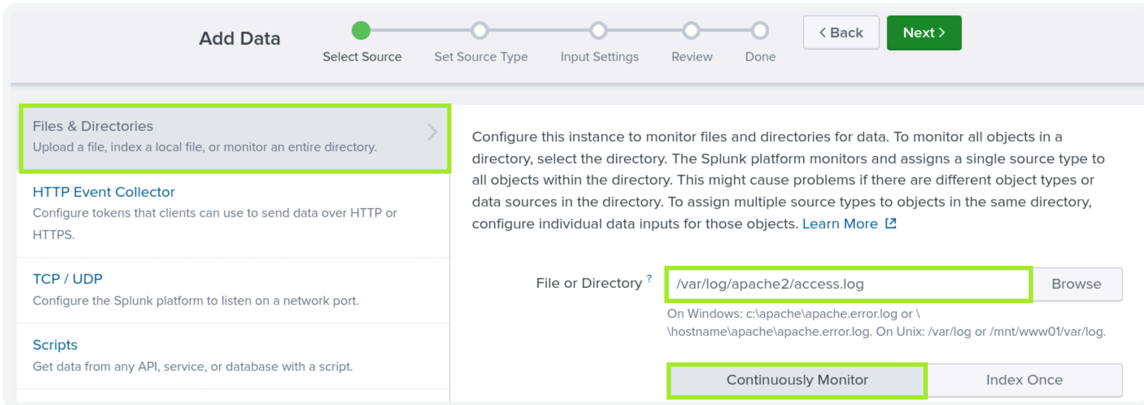
In your Splunk instance, head back to **Settings** and click **Add Data**.



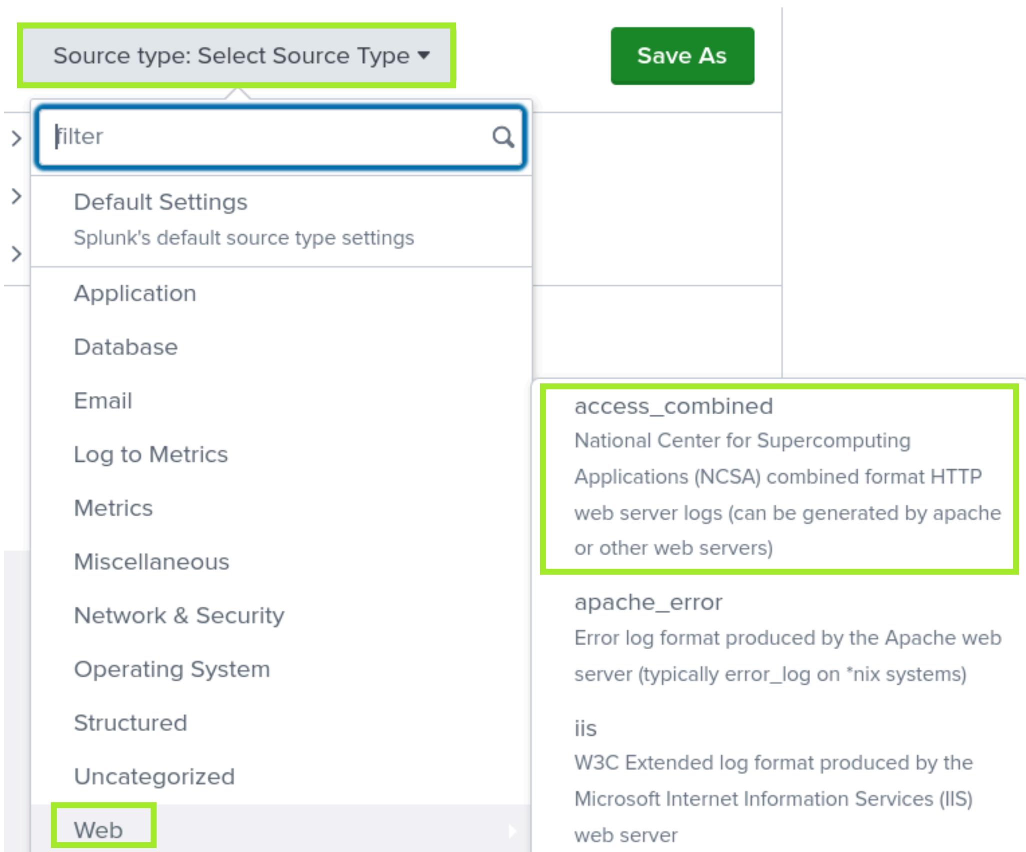

Scroll down and choose the **Monitor** option.



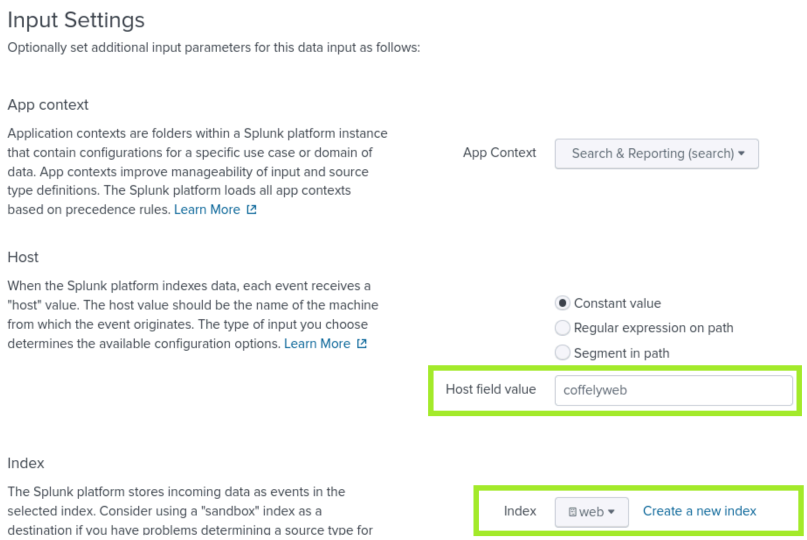

Next, choose **File & Directories** and enter the path of the file we want to monitor: `/var/log/apache2/access.log`. For reference, you can choose to index the file once, but we want to continuously monitor it so that all logs are ingested as they arrive.



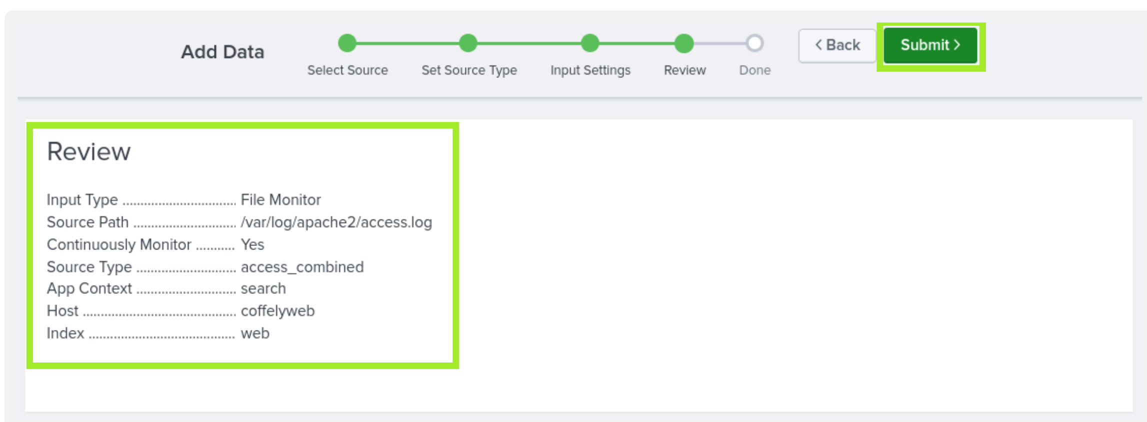

Next, we must select our source type. Click the **Select Source Type** dropdown, select Web, and then choose `access_combined`.



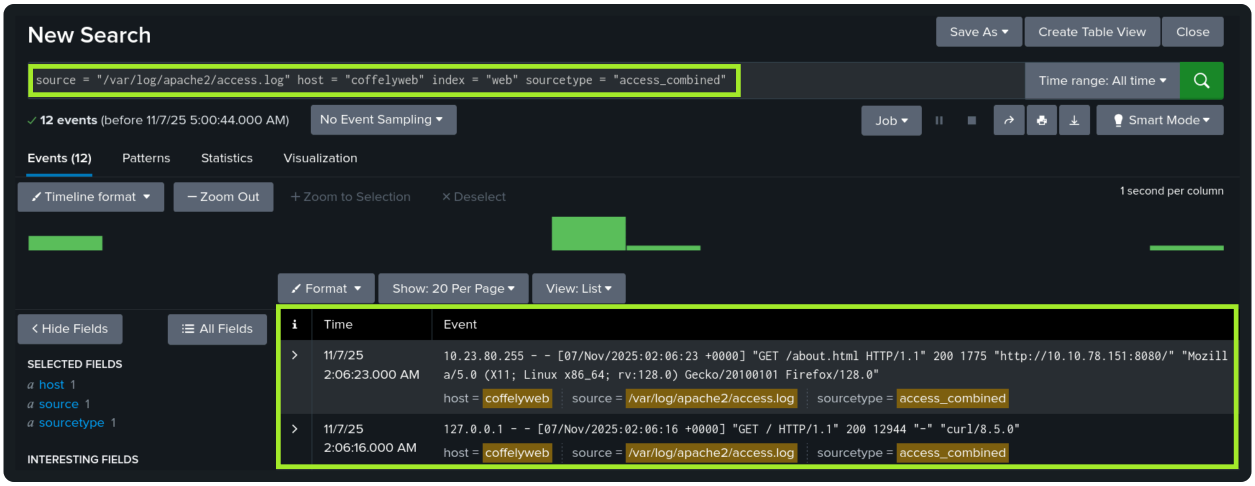

Next, we need to assign a name to the host field value. Let's change it to `coffelyweb`, then create a new index named `web`, just like we did for the previous logs.




Finally, we can review, submit, and start searching through our newly monitored file. You probably won't see any logs at first, so let's head to `http://coffely.thm:8080` and browse around to generate some logs.




Awesome! You have successfully ingested the Apache access logs into Splunk.




### Answer the questions below

> Which sourcetype should be used to parse and categorize Apache access logs correctly?

access_combined

> After browsing the Coffely site, which root field value has the highest count?

images

> What is the full uri_path field value for cappuchino.png?

/images/coffeerecipes/cappuchino.png

> Navigate to http://coffely.thm:8080/secret-flag.html Check out the list of recent orders and find the flag!

THM{best_coffee_in_town!} (visit http://coffely.thm/secret-flag.html; it will display the history logs of the orders made so far.)

## Task 9 - Conclusion

**Splunk Fundamentals:** Installed and configured Splunk and Universal Forwarder, collected OS/web logs, managed Splunk via CLI, and used a deployment server for centralized forwarder management.